

- **Report Exercise: Phenology Modelling**



Applied Geodata Science II

Session 5

Fabian Bernhard

14.10.2025



What is phenology?

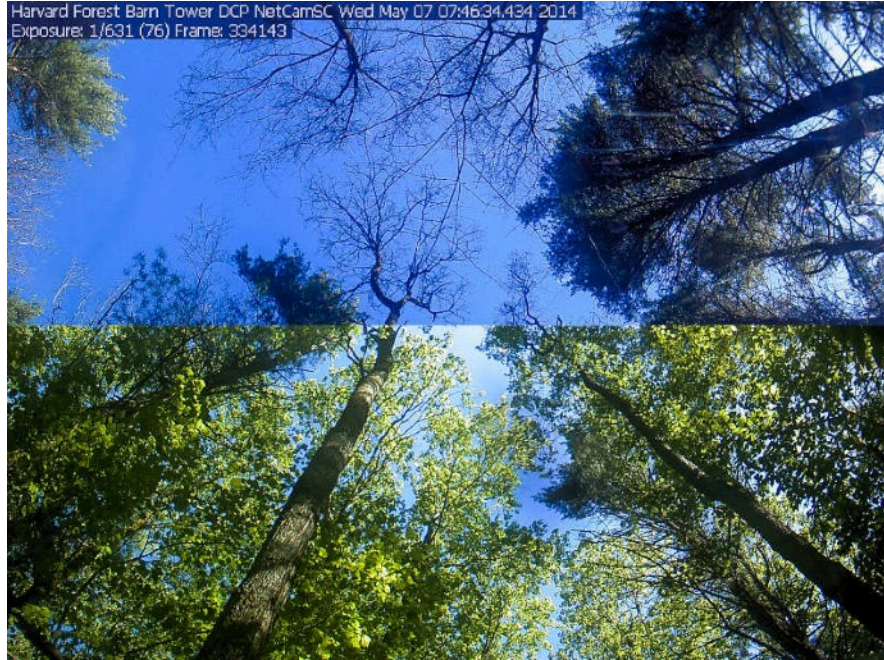


Figure 4.1: Split image of canopy change between winter and spring/summer at Harvard Forest, Petersham, MA, USA

How to observe/measure phenology?

Observational methods:

- Human observation,
e.g. cherry blossoms in Kyoto
- Proximal sensing:
e.g. PhenoCam network
- Satellite based remote sensing:
e.g. MODIS satellites



Techniques to observe phenological changes of forests. (Generated with perplexity.ai)

How to measure phenology?

- Human observation,
e.g. cherry blossoms in Kyoto

INTERNATIONAL JOURNAL OF CLIMATOLOGY
Int. J. Climatol. 28: 905–914 (2008)
Published online 1 August 2007 in Wiley InterScience
(www.interscience.wiley.com) DOI: 10.1002/joc.1594



Phenological data series of cherry tree flowering in Kyoto, Japan, and its application to reconstruction of springtime temperatures since the 9th century

Yasuyuki Aono* and Keiko Kazui

Graduate School of Life and Environmental Sciences, Osaka Prefecture University, Japan

ABSTRACT: Changes in springtime temperature in Kyoto, Japan, since the 9th century were reconstructed, using the phenological data series for cherry tree (*Prunus jamasakura*), deduced from old diaries and chronicles. Phenological data for 732 years was made available by combining data from previous studies. The full-flowering date of cherry trees fluctuates in accordance with temperature conditions during February and March. Full-flowering dates were closely related to the March mean temperature by means of a temperature accumulation index, in which plant growth is considered to be an exponential function of temperature. Calibration enabled accurate estimation of temperatures in the instrumental period, after 1880; the root mean square error (RMSE) of temperature estimates was determined to be within 0.1 °C, after smoothing by local linear regression over time spans of 31 years. The results suggested the existence of four cold periods, 1330–1350, 1520–1550, 1670–1700, and 1825–1830, during which periods the estimated March mean temperature was 4–5 °C, about 3–4 °C lower than the present normal temperature. These cold periods coincided with the less extreme periods, known as the Wolf, Spoerer, Maunder, and Dalton minima, in the long-term solar variation cycle, which has a periodicity of 150–250 years. The sunspot cycle length, a short-term solar variation cycle, was also compared with the temperature estimates, with the result that a time lag of about 15 years was detected in the climatic temperature response to short-term solar variation. Copyright © 2007 Royal Meteorological Society

CLIMATIC RECONSTRUCTION USING THE PHENOLOGY OF CHERRY TREE FLOWERING

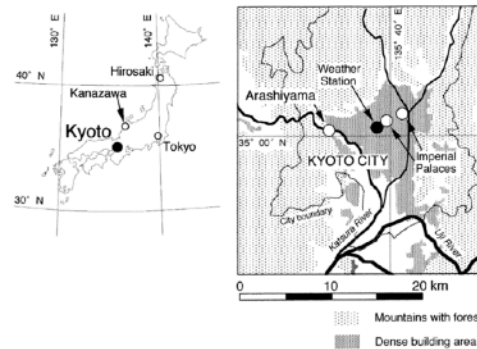
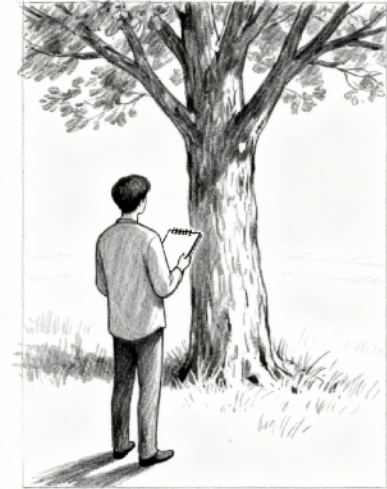


Figure 1. Map of the Kyoto area, and locations of Kyoto and Kanazawa in Japan.



How to measure phenology?

- Human observation,
e.g. cherry blossoms in Kyoto

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Y. AONO AND K. KAZUI

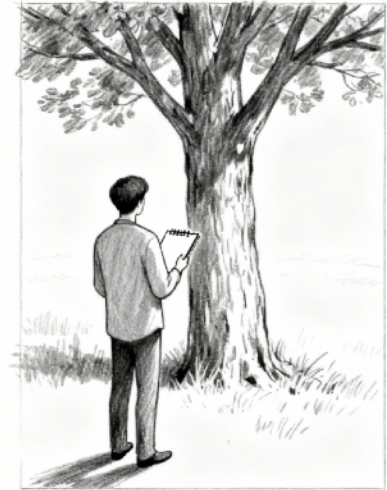
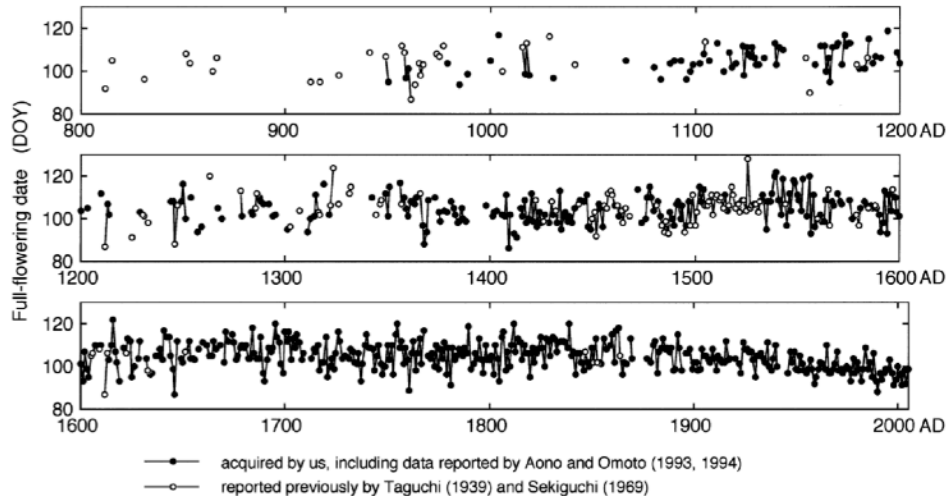
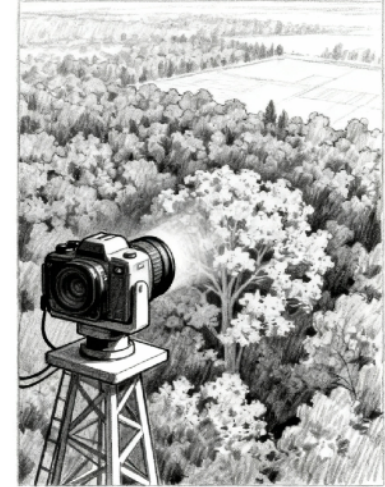
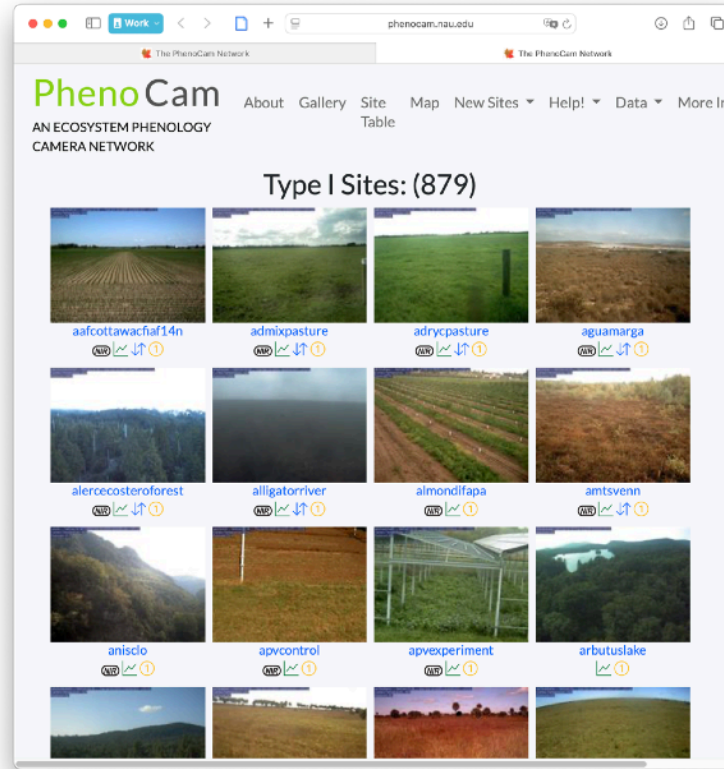


Figure 2. Inter-annual variation of the full-flowering dates of the cherry tree, *P. Jamasakura*, at Kyoto, acquired from old diaries and chronicles.

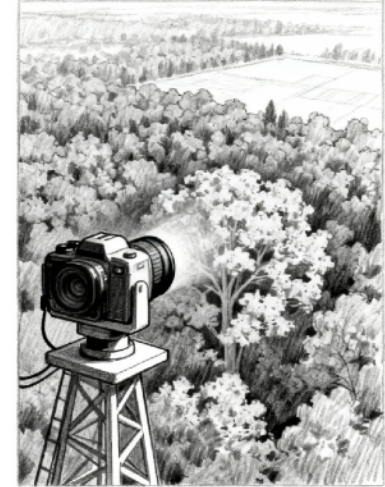
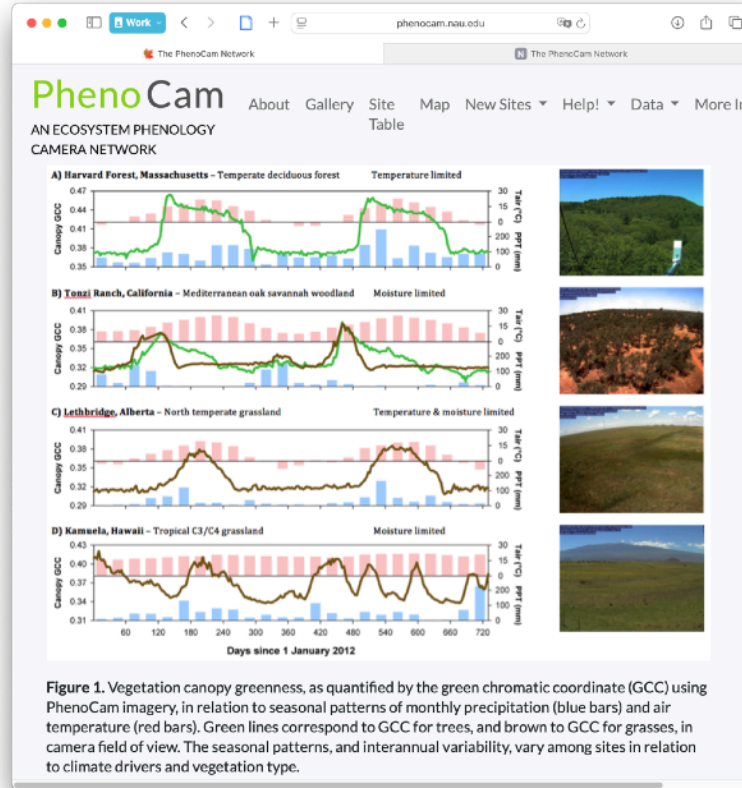
How to measure phenology?

- Proximal sensing:
e.g. PhenoCam network



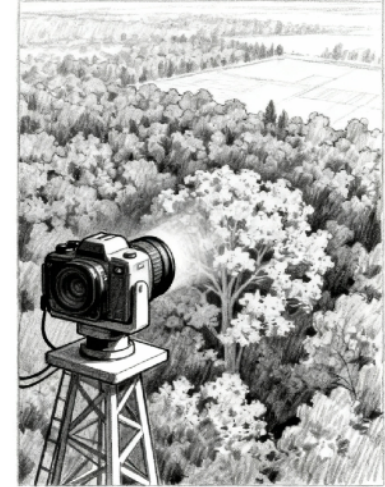
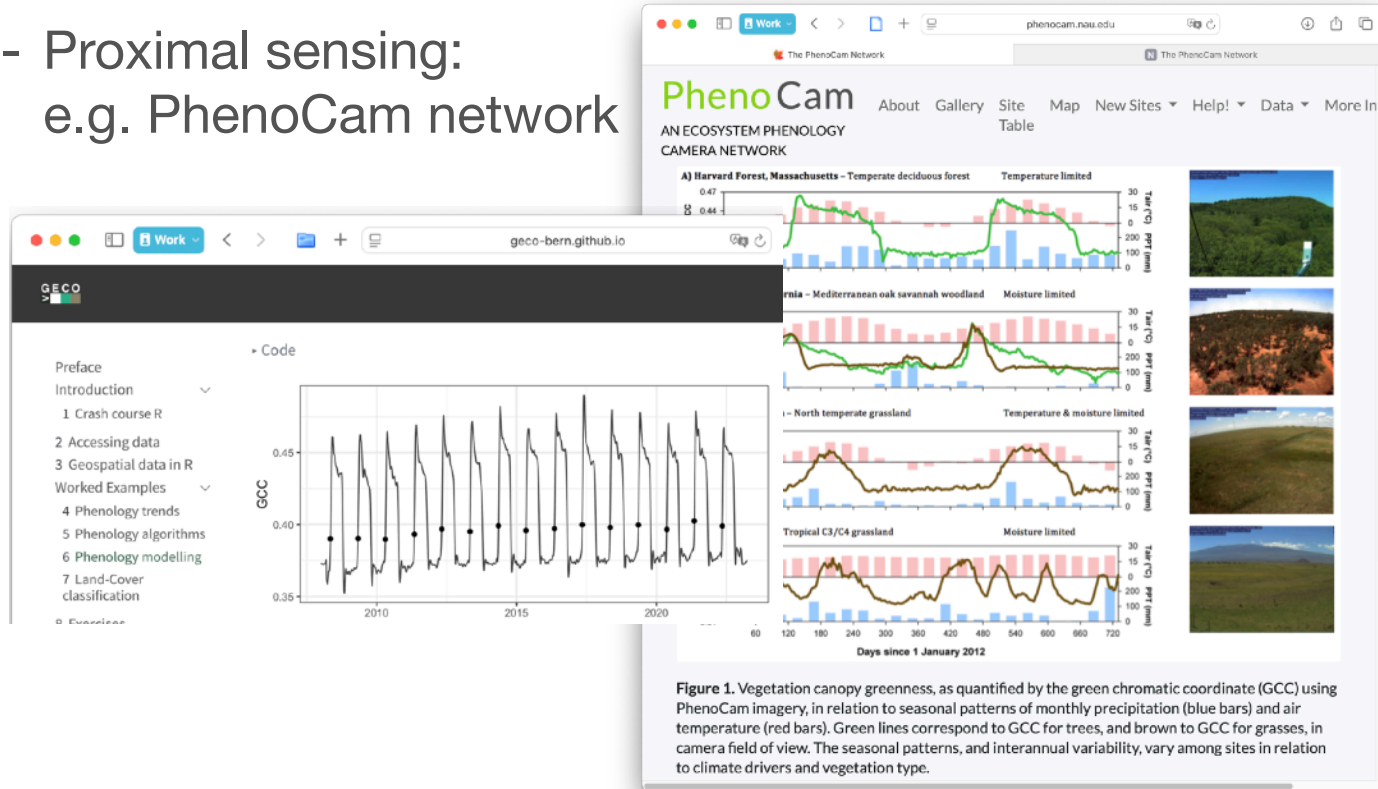
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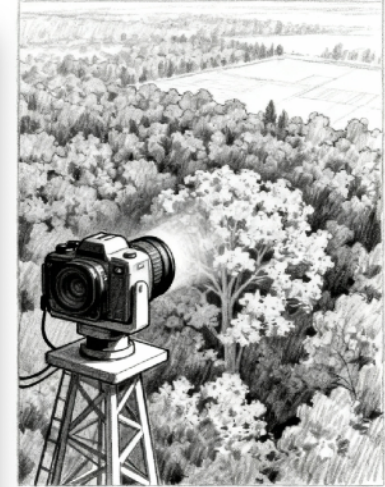
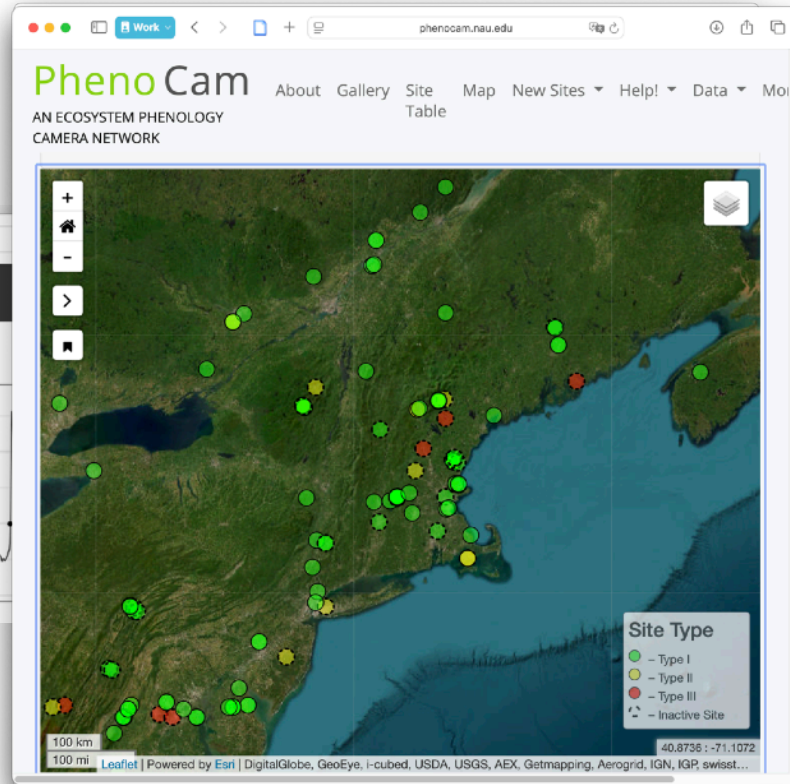
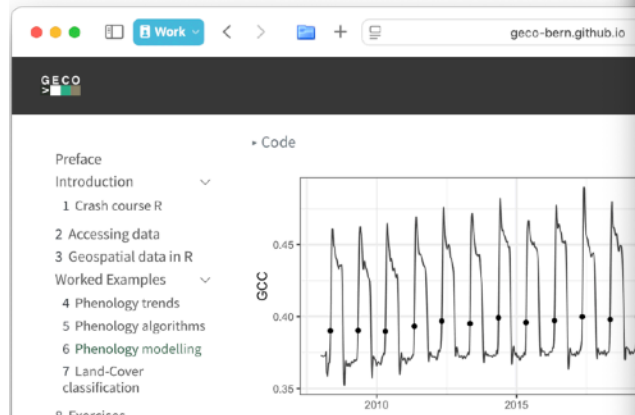
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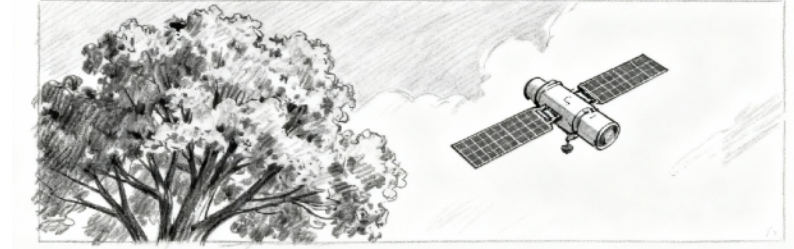
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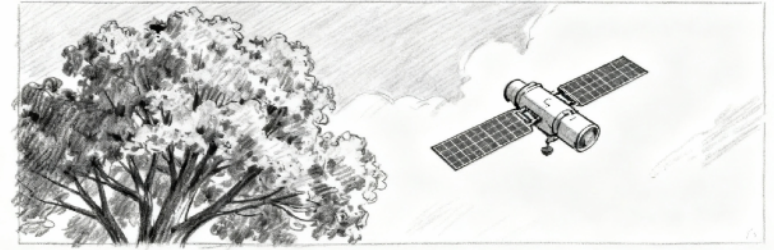
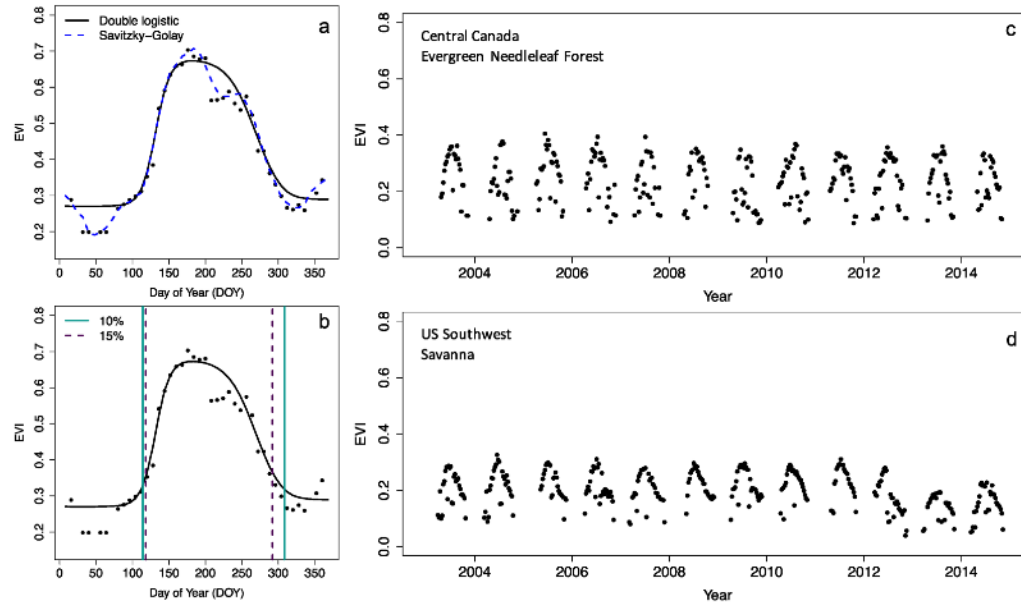


Figure 5.1: Example of a threshold-based phenology algorithm, by Stanimorova et al. 2019

How to measure phenology?

- Satellite based remote sensing:
e.g. MODIS satellites

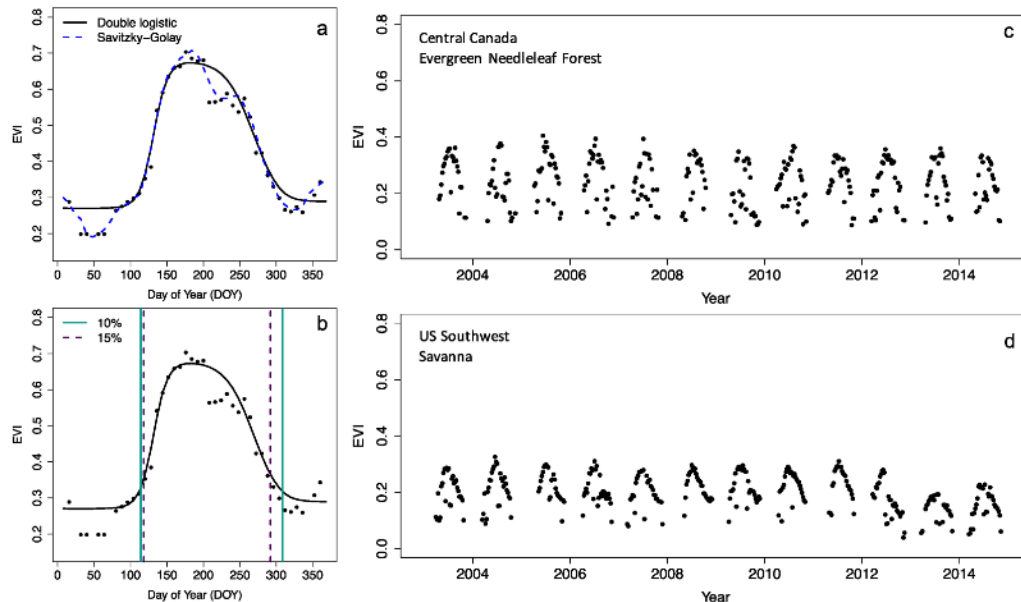
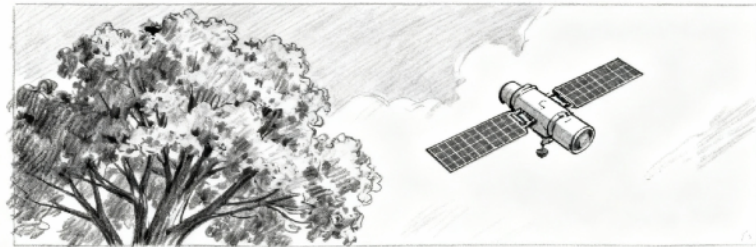


Figure 5.1: Example of a threshold-based phenology algorithm, by Stanimorova et al. 2019



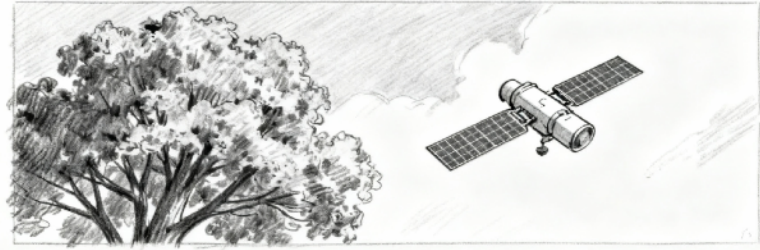
[MODIS land surface phenology product \(MCD12Q2\).](#)

This remote sensing-based data product quantifies land surface phenology and is a good trade-off between data coverage (global) and precision (on a landscape scale).

To access this data, we will use the [MODISTools](#) package. See Chapter [Chapter 2](#)

How to measure phenology?

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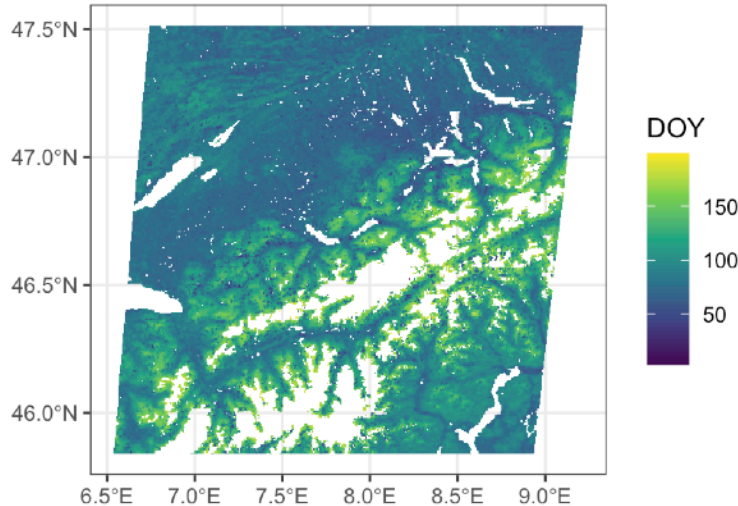
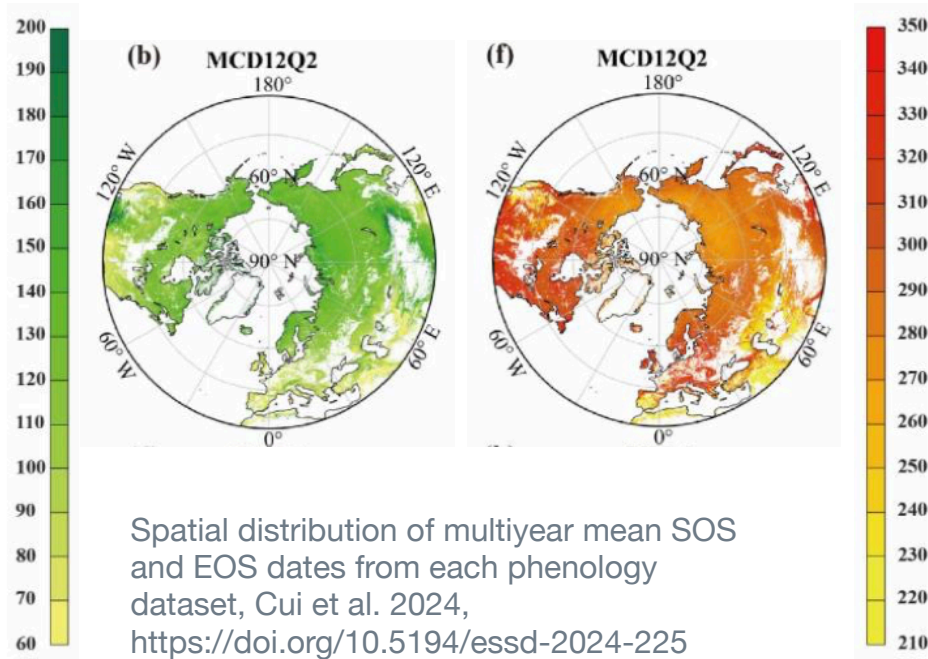
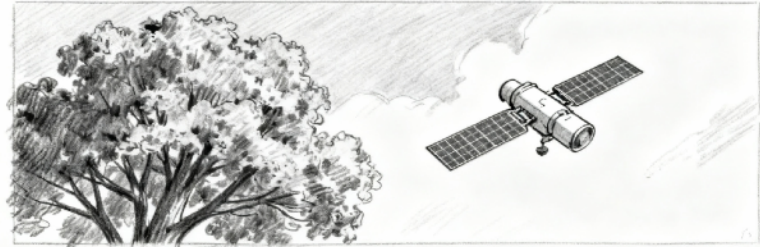


Figure 4.2: MODIS land surface phenology (as DOY for 2012)

How to measure phenology?

- Satellite based remote sensing:
e.g. MODIS satellites



MODIS land surface phenology product (MCD12Q2).

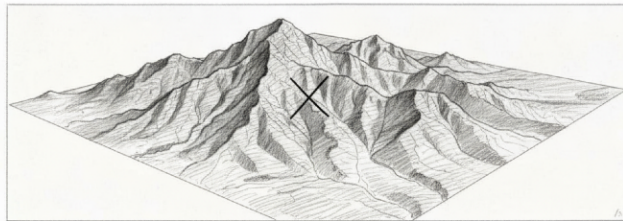
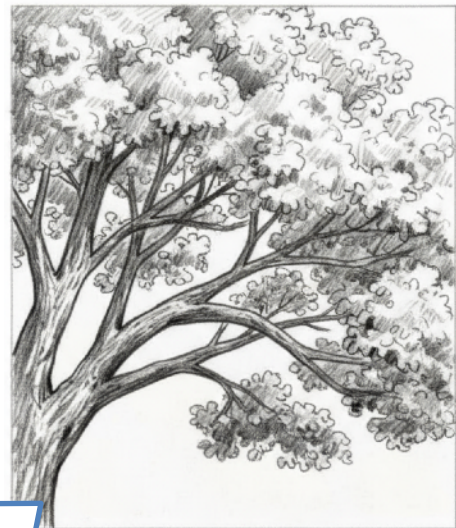
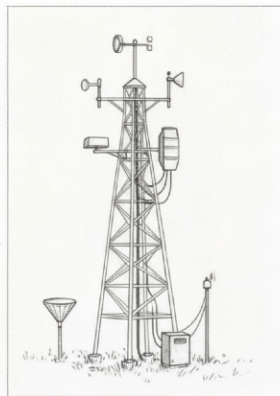
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To access this data, we will use the **MODISTools** package.

Growing Degree Days: to predict leaf-out date

Models can be:

- Statistical black boxes:
e.g. Machine Learning (ML) based
 - Leaf-out (DOY) = $f(\text{Temp.}, \text{Sunshine}, \text{m.a.s.l.}, \dots)$
- Mechanistical, process-based:
e.g. Growing Degree Days
 - Leaf-out (DOY) = $f(\text{Temp.})$

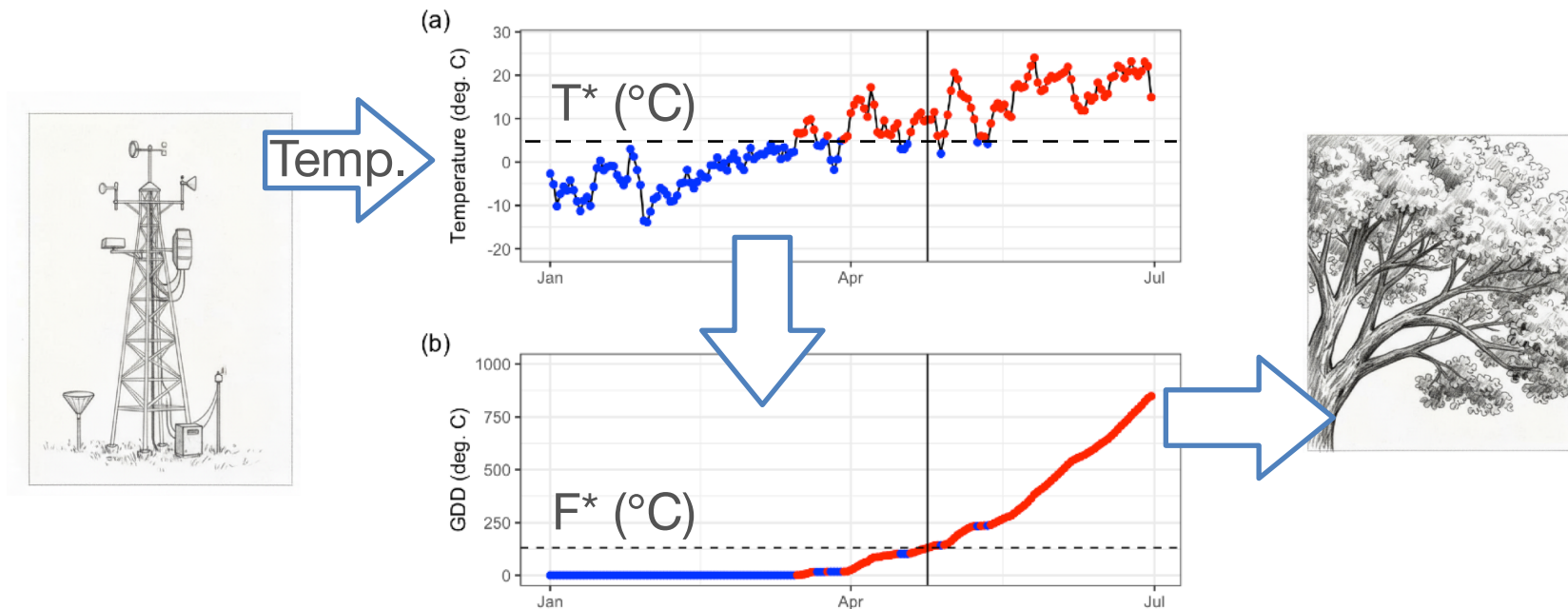


Growing Degree Days: to predict leaf-out date

Leaf-out (DOY) = $f(\text{Temp.}; T^*, F^*)$

Critical T threshold: T^* ($^{\circ}\text{C}$)

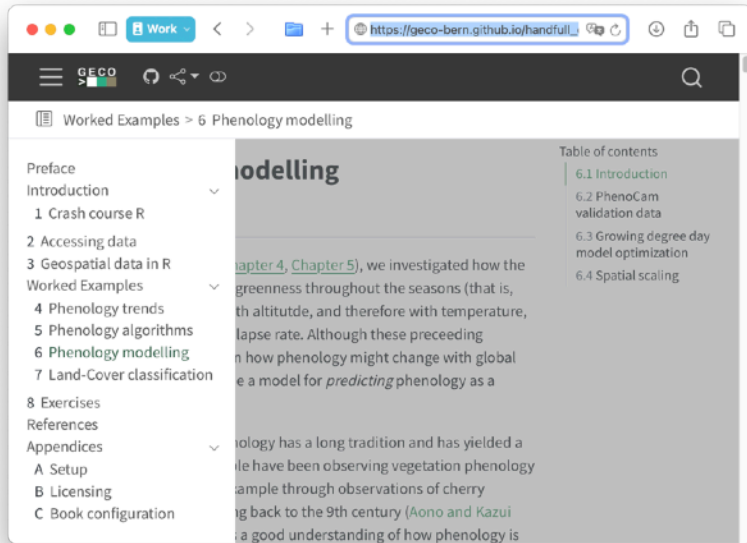
Critical GDD threshold: F^* ($^{\circ}\text{C}$)



Report Exercise

5 Phenology Modelling

14.10.25 - Handful of Pixels Ch.6/ Ex.8.2

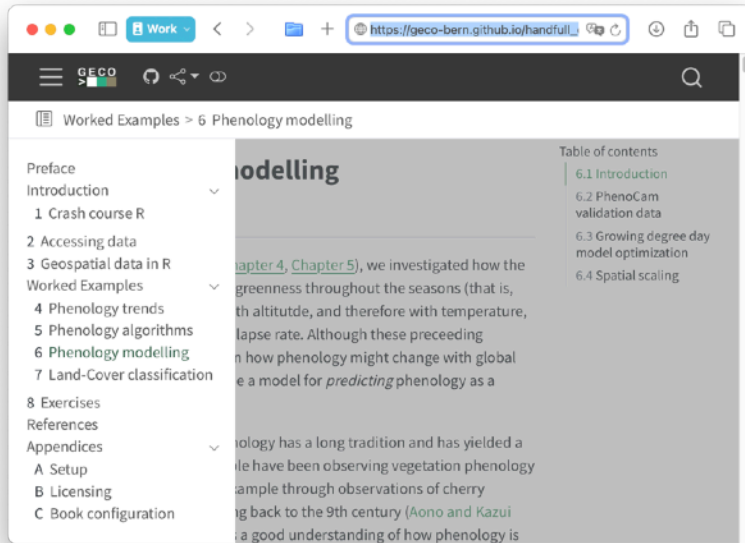


Snapshot of the “Handful of Pixels” online book:
https://geco-bern.github.io/handful_of_pixels/

Report Exercise

5 Phenology Modelling

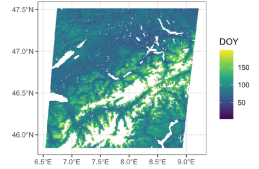
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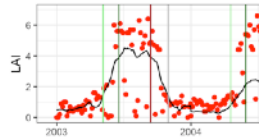
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- Ch. 4: visualize leaf-out dates



- Ch. 5: derive leaf-out dates



- Ch. 6: Mechanistic phenology modelling

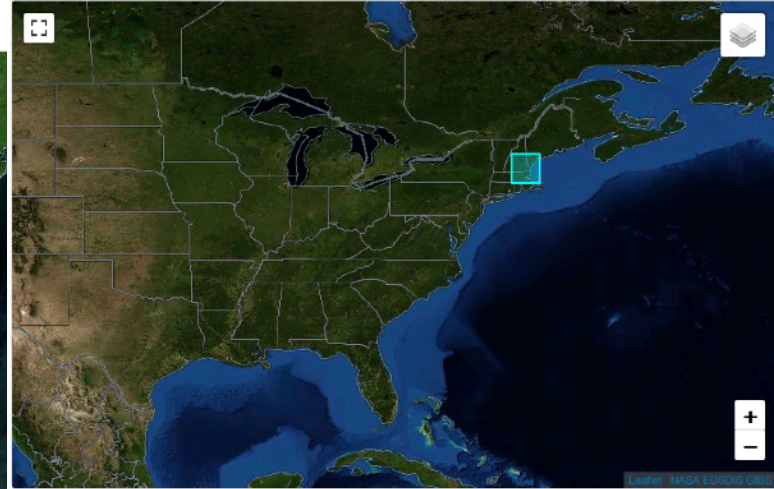
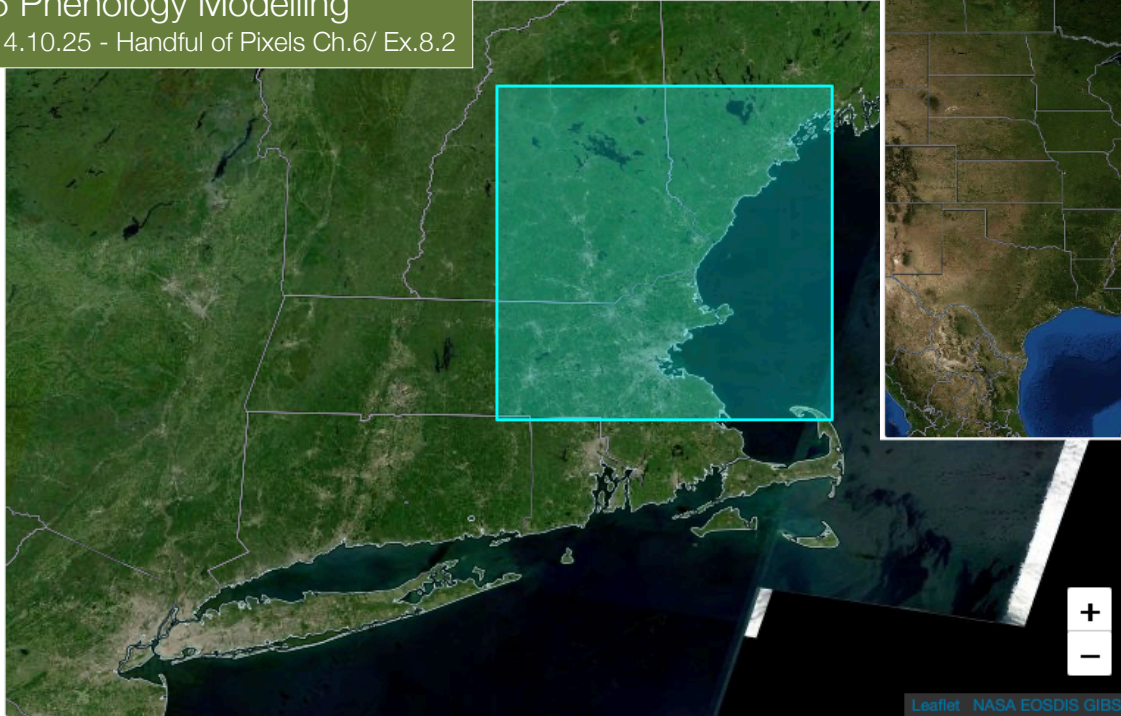


- Ex. 8.2: Improve model from Ch.6

Report Exercise

5 Phenology Modelling

14.10.25 - Handful of Pixels Ch.6/ Ex.8.2



Region of Interest: greater Boston area

Report Exercise

5 Phenology Modelling

14.10.25 - Handful of Pixels Ch.6/ Ex.8.2

8.2 Phenology modelling

- How can you improve the model used to regionally scale the results in [Chapter 6](#)?
 - Provide at least three (3) ways to improve the model used.
- Implement at least one of these methods
- Statistically compare the results with the MODIS MCD12Q2 phenology product
 - compare the data spatially
 - describe why you might or might not see the same patterns
 - consider that 2010 was a 'special' year for the north east of the US

Table of contents

- 8.1 Phenology trends and algorithms
- 8.2 Phenology modelling**
- 8.3 Land-Use and Land-Cover modelling

FINAL REPORTS

- Hand in as reproducible code (*git* repository)
- 2 out of 4 report exercises (separate reports as RMarkdown files)
- Including knitted HTML files
- Deadline for sending us the link by email: 23.12.2025

Report Exercise: Part 1 Model fitting

5 Phenology Modelling

14.10.25 - Handful of Pixels Ch.6/ Ex.8.2

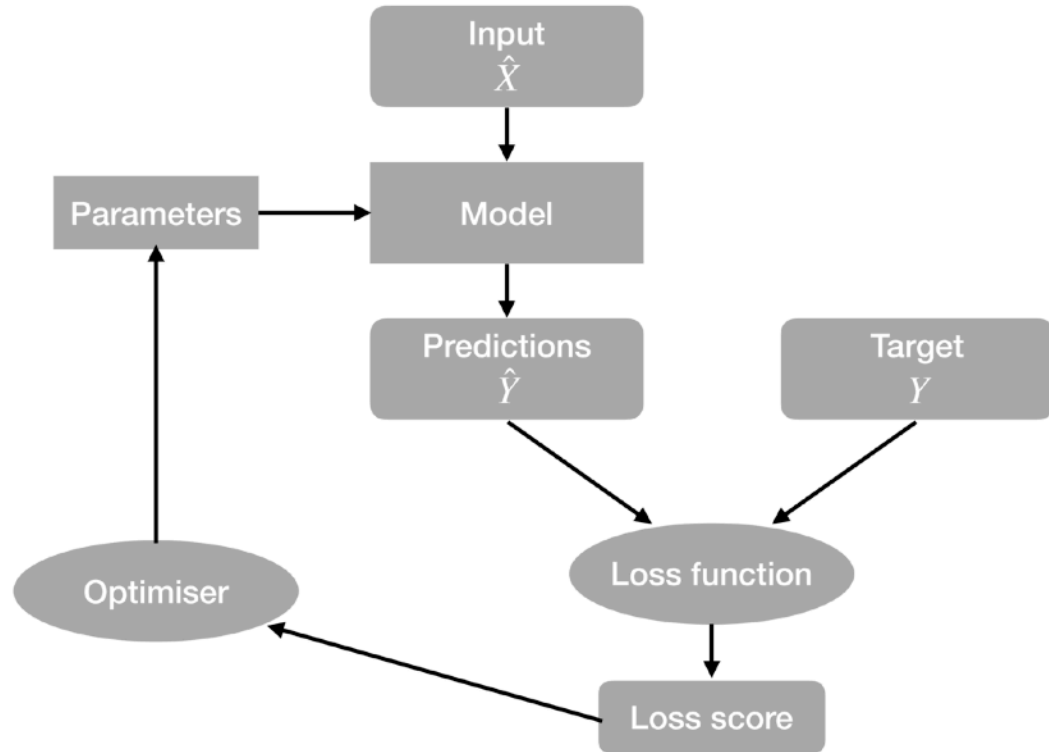


Figure adapted from Chollet & Allaire (2018)
Drawings generated with perplexity.ai

Report Exercise: Part 1 Model fitting

5 Phenology Modelling

14.10.25 - Handful of Pixels Ch.6/ Ex.8.2

Critical T threshold: T^* (°C)
Critical GDD threshold: F^* (°C)

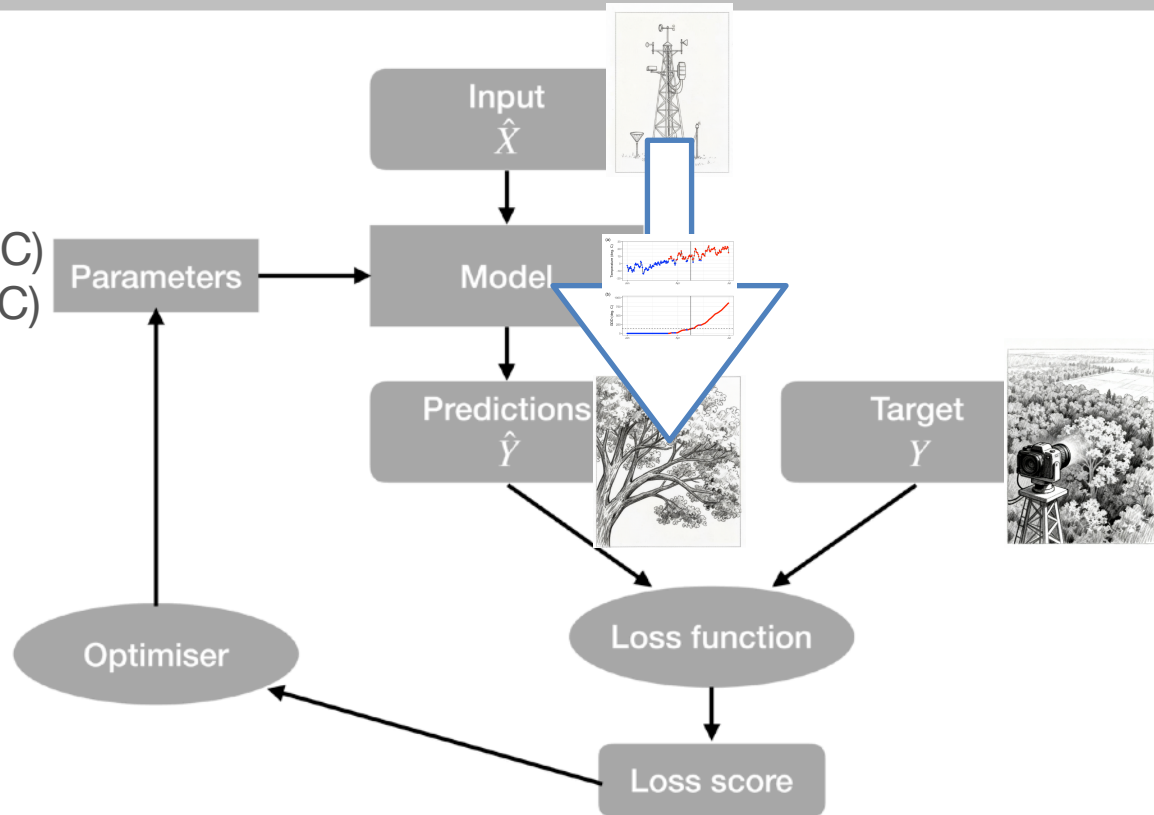
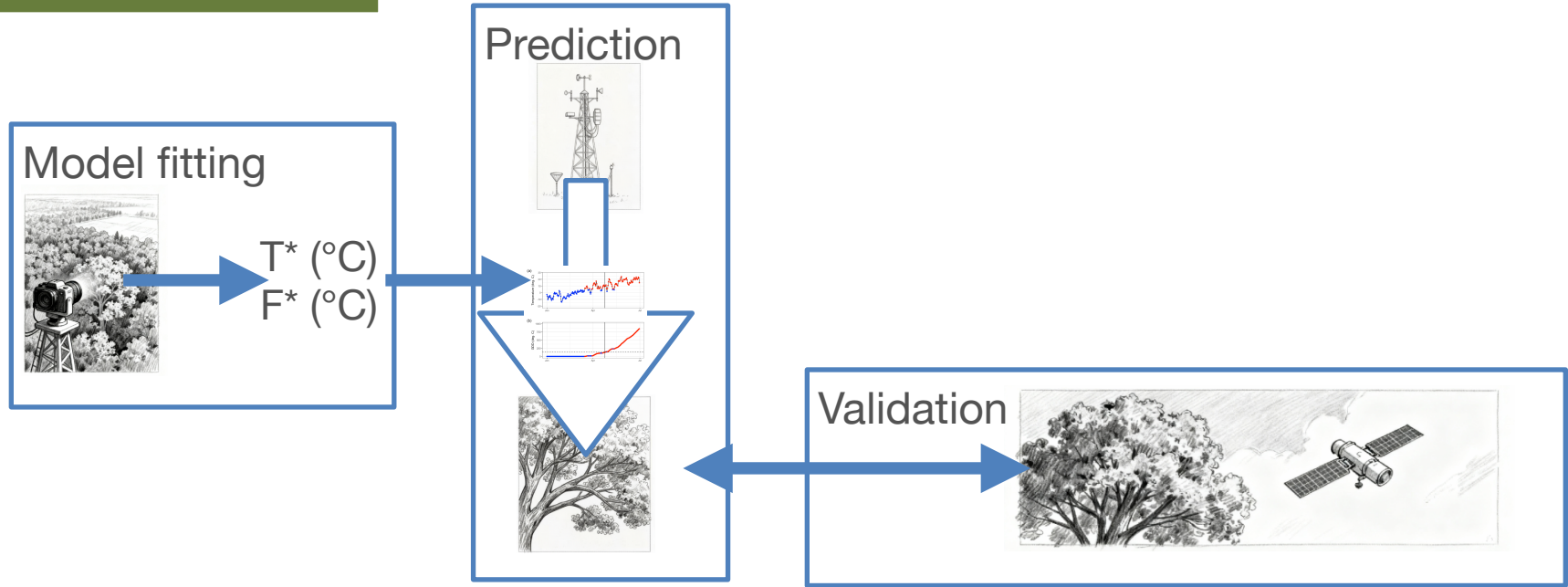


Figure adapted from Chollet & Allaire (2018)
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Report Exercise: Part 2 Regional prediction

5 Phenology Modelling

14.10.25 - Handful of Pixels Ch.6/ Ex.8.2

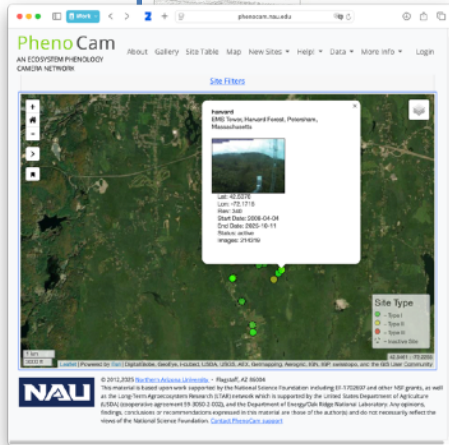


Report Exercise: Part 2 Regional prediction

5 Phenology Modelling
14.10.25 - Handful of Pixels Ch.6/ Ex.8.2

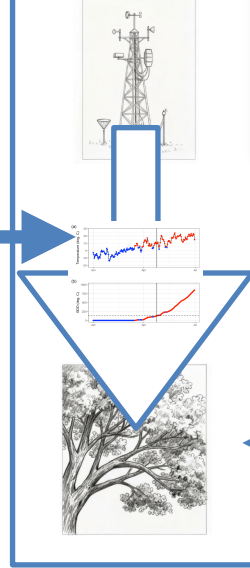
Daily temperature from Daymet

Model fitting

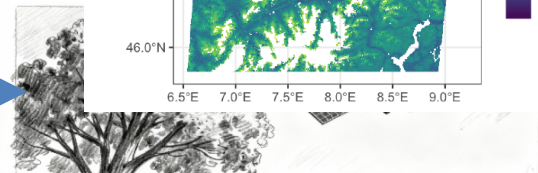


(°C)
(°C)

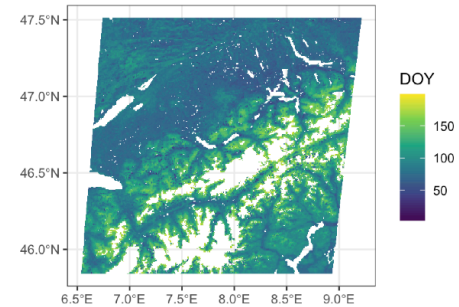
Prediction



Validation



MODIS land surface
phenology (as DOY) MCD12Q2



Harvard Phenocam

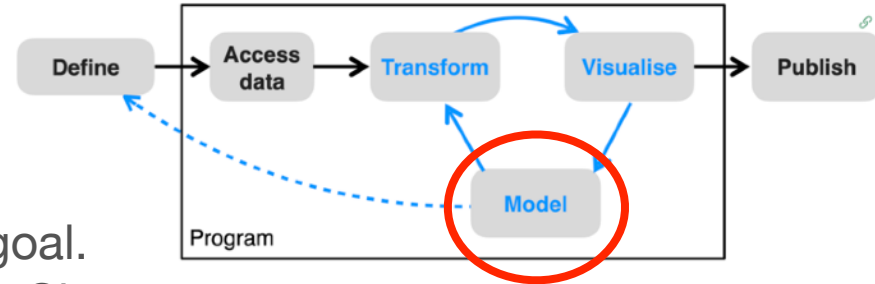
Report Exercise: Tips

5 Phenology Modelling

14.10.25 - Handful of Pixels Ch.6/ Ex.8.2

As usual:

- understand the input data, understand the goal.
- understand the data analysis approach from Ch6
- think about the data science workflow
- use the help in R
(`?download_phenocam`, vignettes: `??appeears` `??phenocamr`)
and data documentations:
 - <https://phenocam.nau.edu/webcam/>
 - <https://www.bu.edu/lcsc/data-documentation/>
 - or ...
- decide yourself which parts of `Handful of Pixels` you need to read to solve the exercise



Data science workflow